DGMS 2019 GEOLOGY • QUESTION 35

Optical Mineralogy: Hexagonal Crystals & The Optic Axis

Topic: Optical Mineralogy & Crystallography | Primary Reference: Dexter Perkins (Mineralogy, 3rd ed., Ch. 5)

Question Statement

Q.35. In hexagonal crystals, there is one and only one direction which shows no double refraction of light. This direction is called

(a) Optic axis
(b) Optic axial plane
(c) Isogyre
(d) Optic normal

a **Optic axis (Correct Answer)**	b Optic axial plane
c Isogyre	d Optic normal

✓ **Correct Answer: (a) Optic axis [In hexagonal and tetragonal crystals, the crystallographic c-axis is the single optic axis along which birefringence is zero].**

Textbook Evidence: Dexter Perkins (Chapter 5)

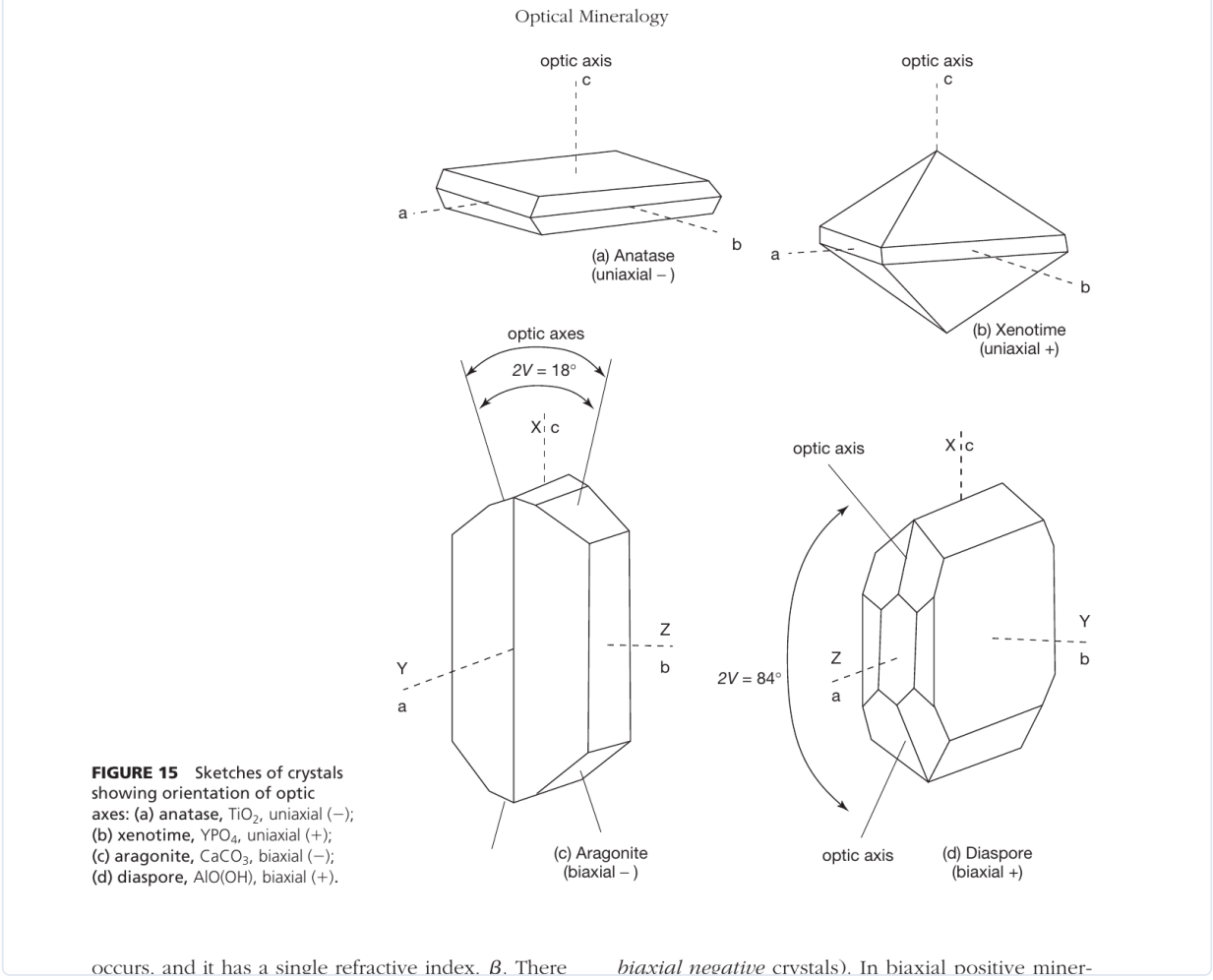


FIGURE 15 Sketches of crystals showing orientation of optic axes: (a) anatase, TiO_2, uniaxial (−); (b) xenotime, YPO_4, uniaxial (+); (c) aragonite, $CaCO_3$, biaxial (−); (d) diaspore, AlO(OH), biaxial (+).

occurs, and it has a single refractive index, β. There *biaxial negative* crystals). In biaxial positive miner-

Figure 5.15: The Uniaxial Optical Indicatrix (Dexter Perkins, *Mineralogy*, 3rd ed., pp. 122–123). In minerals belonging to the hexagonal and tetragonal crystal systems, the optical indicatrix is an ellipsoid of revolution about the crystallographic c-axis. The circular cross-section is perpendicular to c. Light propagating along the c-axis vibrates in the circular section with a constant index of refraction (n_ω). It experiences **zero birefringence ($\Delta n = 0$)** and no double refraction. This unique propagation direction is the **Optic Axis**.

> *"For light traveling parallel to the c-axis, all vibration directions are perpendicular to c and have the same refractive index (n_ω). Because all vibration directions are equivalent, light traveling parallel to c does not split into two rays. It behaves as if it were in an isotropic mineral. The direction parallel to c in tetragonal and hexagonal minerals is called the* **optic axis**. *Because minerals of these systems have only one optic axis, they are called* **uniaxial minerals**."
>
> **— Dexter Perkins, *Mineralogy*, Chapter 5 (Optical Mineralogy), p. 122**

Comprehensive Option Analysis

(a) Optic axis **[CORRECT]**

Definition: An *optic axis* is a direction of light propagation in an anisotropic crystal along which there is **no double refraction (no birefringence)**. Light traveling along this direction travels with uniform wave velocity regardless of vibration direction, effectively experiencing an isotropic section.

- In **hexagonal (including trigonal)** and **tetragonal** crystal systems, the high symmetry around the principal rotation axis makes the c-axis the single optic axis (hence *uniaxial*).
- Under crossed polars in a petrographic microscope, a grain cut strictly perpendicular to the optic axis remains permanently extinct throughout 360° stage rotation, and conoscopically exhibits a classic centered uniaxial cross interference figure.

(b) Optic axial plane (OAP) **[INCORRECT]**

The *Optic Axial Plane (OAP)* is a geometric plane found **only in biaxial crystals** (orthorhombic, monoclinic, triclinic). It is the plane that contains both optic axes, the acute bisectrix (Bxa), and the obtuse bisectrix (Bxo). Uniaxial (hexagonal) crystals have only *one* optic axis, so no optic axial plane exists in the hexagonal system.

(c) Isogyre **[INCORRECT]**

An *isogyre* is an optical interference feature, not a crystallographic direction. When observing a crystal conoscopically (using the Bertrand lens with convergent polarized light), the dark bands or cross formed where light vibration directions coincide with the vibration directions of the polarizer and analyzer are called isogyres.

(d) Optic normal **[INCORRECT]**

The *Optic Normal* is the vibration direction Y (n_β) in a **biaxial crystal**. It is oriented exactly perpendicular to the optic axial plane ($Y \perp \text{OAP}$). It has no meaning in the hexagonal (uniaxial) system.

Optical Symmetry Systematics Across Crystal Systems

Optical Class	Crystal Systems	Indicatrix Geometry	Number of Optic Axes	Optic Orientation
Isotropic	Isometric (Cubic), Amorphous	Sphere ($n_x = n_y = n_z$)	Infinite (no unique axis)	Every direction is non-birefringent
Uniaxial	**Hexagonal, Trigonal, Tetragonal**	**Ellipsoid of Revolution (n_ω, n_ϵ)**	**1 (One unique axis)**	**Optic axis $\parallel c$-crystallographic axis**
Biaxial	Orthorhombic, Monoclinic, Triclinic	Triaxial Ellipsoid ($n_\alpha, n_\beta, n_\gamma$)	2 (Two optic axes)	Lie in Optic Axial Plane (OAP); $Y \perp \text{OAP}$ (Optic Normal)

Key Takeaway & Exam Strategy

Memory Matrix for Competitive Exams:

- **Hexagonal** / **Tetragonal:** Uniaxial $\rightarrow$ **1 Optic Axis** coincident with the c-axis.
- **Orthorhombic** / **Monoclinic** / **Triclinic:** Biaxial $\rightarrow$ **2 Optic Axes**, separated by $2V$, lying in the **Optic Axial Plane (OAP)**, with **Optic Normal (Y)** perpendicular to OAP.
- Direction of zero double refraction $\equiv$ **Optic Axis**.